

Taking an ASD facial-emotion benchmark to a Q1 journal

A working session on evaluation protocol, statistical reporting, figure design and model selection for a six-class autism facial emotion recognition study.

Participant	Md. Ashiqur Rahman
Assistant	Claude (Anthropic)
Date	30 August 2026
Exchanges	8 questions
Source material	Research Overview and Progress.pdf — 2,003 images, 4 merged sources, 6 emotion classes
Files produced	2 Word documents · 4 Python modules · 17 figures · 7 result tables · 3 web pages

WHAT THIS TRANSCRIPT CONTAINS

The questions asked and the answers given, in full. It does not reproduce the intermediate work behind those answers — literature searches, code written and executed, figures rendered and inspected, or the verification runs. Files delivered during each exchange are listed at the end of that exchange; they are separate downloads and are not embedded here.

Attached: Research Overview and Progress.pdf

Hey claude. I want to submit it in a Q1 journal. Give me a full overall pipeline how to publish it in a Q1 journal. How to organize my result section and also how to organize my full paper.

During Baseline what i need to check and what analysis i need to be done for a high impact factor Q1 journal.

And also give me suggestion how to implement baseline model step by step. Give me a baseline code as well.

CLAUDE

CLARIFICATIONS REQUESTED

Target venue: biomedical / health informatics · Deliverable: guide page plus a runnable script · Code scope: publication-grade rewrite

The short version of what I found in your progress report.

Your training harness is good — better than most submissions in this space. But three things would sink the paper at JBHI or CBM before anyone evaluates your architecture:

1. **Your splits aren't subject-independent.** Merged face corpora have many frames per child. Under a 15% image-level split, a child with just 3 frames has a 99.7% chance of appearing in training too. Every number in your table is inflated by an unknown amount.
2. **Your headline claim isn't statistically supported.** VGG-16, Swin Base and Inception V3 span 0.0005 accuracy. A paired McNemar test at $n=2003$ can only resolve gaps above ~ 0.028 . "VGG-16 and Swin Base achieved the highest performance" is a ranking of noise. Attach Wilson intervals to your own numbers and 0.7201 becomes 0.700–0.739 — and that's before correcting for within-subject correlation, which widens it by roughly another 80%.
3. **Four merged datasets = four cameras, four label protocols.** You need the source-probe and leave-one-dataset-out experiments to show how much of your accuracy is emotion and how much is dataset fingerprint.

On novelty: ten fine-tuned backbones won't clear Q1 on their own. Your own motivation contains the strongest hook — you wrote that ASD facial movement is *less symmetrical*. Build that in: hemiface-split encoders fused by cross-attention, with an asymmetry auxiliary head. The ablation writes itself, and it produces a clinical secondary finding independent of accuracy. It also resolves a contradiction sitting in your current code — you're using horizontal flip augmentation, which imposes exactly the symmetry prior your hypothesis rejects.

On the code: `build_manifest.py` recovers the identity groups you're missing (FaceNet embeddings + agglomerative clustering) and catches near-duplicates your MD5 dedup missed. `asd_fer_baseline.py` runs grouped CV against a locked test set over multiple seeds, saves per-image probabilities, then computes every statistic offline from those files — so when a reviewer asks for a metric you didn't anticipate, you produce it that afternoon instead of re-running 150 GPU jobs.

I ran the analysis half end-to-end on synthetic predictions: tables and figures generate correctly, the cluster bootstrap widens 53% under realistic within-subject correlation, McNemar returns $p=1$ for identical models, and temperature scaling drops ECE from 0.095 to 0.038. Two real bugs surfaced during that check and are fixed.

One prediction worth bracing for: when you re-run under the honest protocol, your macro-F1 will drop. That drop is not a setback — quantified properly, it's publishable evidence about how much reported ASD-FER performance across the field is leakage, and it's your fallback paper if the hybrid ends up tied with the baselines.

DELIVERED IN THIS TURN

- Artifact: ASD-FER Q1 Submission Protocol
- build_manifest.py
- asd_fer_baseline.py

MD. ASHIQUR RAHMAN

2

Also add different curves, Train vs Validation curve, Confusion Matrix, ROC-AUC curve, And all other relevant curves.

And what your code gives actually the output.

CLAUDE

The gallery page shows every output, generated by actually running the code — not mocked up. I built a simulated corpus matched to your scale (2,003 images, 543 subject groups, 10:1 imbalance, ~0.72 accuracy) so the figures look like what you'll get.

18 figures, covering everything you asked for

Train vs validation (loss and macro-F1, mean over folds with a fold min-max band and the selected epoch marked), per-fold curves, seed stability, confusion matrix (row-normalised + raw counts), a confusion grid across all backbones, ROC per emotion, macro/micro ROC across models, precision-recall per emotion, calibration, risk-coverage, per-class F1 with intervals, a forest comparison, a significance matrix, error agreement, subgroup-by-source, LODO, and dataset composition. An 18th (F06) draws the recall difference between subject-independent and image-level protocols — that's your leakage figure, and it needs two runs.

Four things I changed on purpose, because the obvious version would have been wrong

Loss and F1 are in *separate panels*, never one panel with two y-axes. Where the two lines cross on a dual-axis plot is an artefact of the scaling you picked, and readers over-read it every time.

Six emotion classes are drawn as six small panels, never six overlaid coloured curves. Six hues can't be separated by a colour-blind reader, and they're indistinguishable in greyscale print. Model overlays are capped at three, with dash patterns carrying identity.

The ROC band is a *real* bootstrap band — resampled at each FPR on a grid. My first version shaded the area under the curve and captioned it as a confidence interval. That's a reporting error a statistical reviewer would write up, so I replaced it.

Reliability plots drop bins holding fewer than ten images. One sparse bin produces a wild accuracy estimate that reads as a calibration failure when it's just noise. ECE still uses every image.

On what the numbers say

Look at T3 in the gallery. swin_base beats vgg16 by 0.005 macro-F1, Holm-adjusted $p = 0.249$. They disagree on 365 images and split them 194/171. That's the row that stops you writing "outperformed." Meanwhile T5 gives you the sentence worth putting in the abstract: abstaining on the least-confident 20% lifts accuracy from 0.694 to 0.883 — and the best-calibrated model isn't the most accurate one, which for a clinical tool is a real decision rather than a footnote.

`asd_fer_baseline.py` is updated and now logs validation loss and train macro-F1 per epoch (your old history only had train loss and val F1, so a train-vs-val plot wasn't possible), and calls the figure module automatically at the end of `--analyze-only`.

DELIVERED IN THIS TURN

- Artifact: ASD-FER Output Gallery
- `asd_fer_figures.py`
- `asd_fer_baseline.py` (updated)
- `make_demo_run.py`
- `asd_fer_demo_outputs.zip` — 17 figures at 300 dpi + PDFs + CSV tables

MD. ASHIQUR RAHMAN

3

What do you think which of the deep learning and transformer based need to explore on this dataset?

CLAUDE

First, something that changes your positioning

A 2025 paper in *Applied Sciences* used **FERAC** — one of your four sources — and benchmarked ResNet152V2, DenseNet201, InceptionV3, InceptionResNetV2, ViT, DeiT, CvT and Swin on it. Swin won at 0.800 accuracy / 0.789 F1. They also report their CNNs failing entirely on fear, the same failure your confusion matrix shows. That's close to the experiment you've run, on a subset of your data. It doesn't sink anything, but the benchmark is now your control condition and needs a citation. Search your other three sources by name before you write more.

On the actual question

At ~1,700 development images, architecture isn't the lever. Your ten backbones already sit inside each other's intervals; an eleventh will too. Two things move the number:

What the weights already know. ImageNet pretraining knows cats and cars — nothing in it knows what a raised brow is. Insert a middle stage: ImageNet → AffectNet → your ASD corpus. A 2026 cross-scenario robustness study found AffectNet-trained models generalised markedly better than RAF-DB-trained ones, and that the *training distribution* mattered more than architecture. This is the only item that produces a finding rather than a leaderboard row. Request AffectNet access today; approval is slow.

How much of the model you let the data move. Full fine-tuning 85M parameters on 1,700 images is memorisation with extra steps. Run all four regimes on your top two backbones — full fine-tune, LoRA (0.49M trainable, 0.56%), a multi-depth

frozen probe (1.19M), and a linear probe (0.005M). If the 0.5M LoRA matches the 85M fine-tune, that's a publishable observation about scarce clinical image data, for the cost of one extra table column.

Frozen DINOv3, probed at intermediate depth. It carries 83% region-level facial correspondence with no face-specific training. The detail people miss: correspondence peaks around block 18 of 24 and *degrades* by the final block, so a standard last-layer linear probe throws away exactly the part that distinguishes a brow from a cheek. This is also what makes your hemiface idea implementable — dense semantic correspondence is what lets you align left brow to right brow across different children. Without it, hemiface comparison is just cropping.

CoAtNet, MaxViT and CAFormer are mandatory, not optional. You're proposing to design a hybrid CNN-Transformer. Neither VGG-16 nor Swin is a hybrid, so nothing in your current table answers “why not use an existing one?” If those three aren't there, that question ends the review.

What to skip: vision-language models (the 2026 study found SmolVLM, PaliGemma, BLIP-2 and FaceLLM all skew hard to happiness with near-zero F1 on negative emotions — precisely your rare classes), CLIP/SigLIP as anything but a negative control, and larger versions of what you have.

One cheap idea worth doing in week 1: extract action units with OpenFace plus explicit hemiface asymmetry statistics, and fit gradient boosting. If asymmetry carries signal, you have evidence for the architecture *before* spending two months building it. If it carries none, you've saved those months.

Two bugs surfaced while verifying the code. `features_only` was truncating a 12-block ViT to its first three blocks — a 22M stump probing edge filters, running without error. And LoRA found nothing to adapt in CoAtNet, whose attention projections are 1×1 convolutions rather than Linear layers. Both fixed; every model in the registry was instantiated and forward-passed in every mode it claims to support.

DELIVERED IN THIS TURN

- Artifact: ASD-FER Model Shortlist
- `asd_fer_zoo.py`

MD. ASHIQUR RAHMAN

4

Give me the model name only. Transformer based model and Deep learning model both.

CLAUDE

All names verified against timm 1.0.29.

CNN (deep learning)

Already have: VGG-16 · ResNet-50 · DenseNet-121 · Inception V3 · EfficientNet-B0 · MobileNetV2

ADD	TIMM NAME
ConvNeXt-V2 Base	convnextv2_base.fcmae_ft_in22k_in1k
ConvNeXt-Base (DINOv3)	convnext_base.dinov3_lvd1689m

ADD	TIMM NAME
DenseNet-201	densenet201
Inception-ResNet-V2	inception_resnet_v2
SE-ResNeXt-50	seresnext50_32x4d
RegNetY-3.2GF	regnety_032

Transformer

Already have: ViT-Base · Swin-Base · Swin-Tiny · DeiT-Small

ADD	TIMM NAME
DINOv3 ViT-B/16	vit_base_patch16_dinov3.lvd1689m
DINOv3 ViT-L/16	vit_large_patch16_dinov3.lvd1689m
DINOv2 ViT-B/14	vit_base_patch14_dinov2.lvd142m
EVA-02 Base	eva02_base_patch14_224.mim_in22k
DeiT-III Base	deit3_base_patch16_224.fb_in22k_ft_in1k
Swin-V2 Base	swinv2_base_window12to16_192to256.ms_in22k_ft_in1k
BEiT-3 Base	beit3_base_patch16_224.in22k_ft_in1k
Hiera Base	hiera_base_224.mae_in1k_ft_in1k
SigLIP ViT-B/16	vit_base_patch16_siglip_224.webli

Hybrid CNN-Transformer

MODEL	TIMM NAME
CoAtNet-2	coatnet_rmlp_2_rw_224.sw_in12k_ft_in1k
MaxViT-Tiny	maxvit_tiny_tf_224.in1k
MaxViT-Small	maxvit_small_tf_224.in1k
CAFormer-S36	caformer_s36.sail_in22k_ft_in1k
ConvFormer-S36	convformer_s36.sail_in22k_ft_in1k
MobileViT-V2	mobilevitv2_100.cvnets_in1k
EfficientFormer-V2-S2	efficientformerv2_s2.snap_dist_in1k
FastViT-MA36	fastvit_ma36.apple_dist_in1k

FER-specific (external checkpoints)

POSTER++ · DAN · APViT · EmoNet · ResMaskNet · SwinFace · LA-Net

Face encoders (external)

ArcFace-R100 · FaRL · FSFM · MARLIN

Two notes: SigLIP goes in as a **negative control**, not a candidate. And the four hybrids are the ones your proposed architecture has to beat — without them in the table, the hybrid contribution has no control condition.

MD. ASHIQUR RAHMAN

5

Give them in a downloadable doc file.

CLAUDE

Word document delivered — `ASD_FER_Model_Shortlist.docx`, three pages.

It covers the five groups (CNN, transformer, hybrid, FER-specialised, face encoders) with exact timm identifiers in a copy-pasteable column. Rows already in your benchmark are shaded so you can see at a glance what's new. The priority order and the “not worth exploring” list are at the end.

I rendered it to PDF and checked every page — tables fit the margins, no overflow, monospace identifiers line up.

DELIVERED IN THIS TURN

- `ASD_FER_Model_Shortlist.docx`

MD. ASHIQUR RAHMAN

6

Make a doc file for ASD FER Q1 submission protocol

CLAUDE

`ASD_FER_Q1_Submission_Protocol.docx` — 15 pages, matching the online version, with the FERAC prior-art finding folded into section 2.

I made the contents list static rather than a Word TOC field. A field renders blank until someone presses F9, and stays blank in any PDF exported before that — so I extracted the real page numbers from a first render and baked them in. It's correct the moment you open it, and in any PDF or Google Docs copy.

Checked all 15 pages: callouts keep their accent bar and tint, tables repeat their header rows across page breaks, and the footer carries page numbers.

DELIVERED IN THIS TURN

- `ASD_FER_Q1_Submission_Protocol.docx`

How I share this chat with others

CLAUDE

Two separate things to share, and they don't travel together.

The chat itself. Click **Share** in the upper right, then **Share** in the pop-out to generate a link. On Free/Pro/Max anyone with the link can open it; on Team/Enterprise only people in your organisation. To turn it off, open the same menu and switch the visibility dropdown from Public to Private.

Three caveats that matter here:

- **Files aren't included.** Your uploaded PDF, and every file I sent you — the two Word documents, the four Python scripts, the figures zip — stay private. A collaborator opening the link sees the conversation but can't download any of them. Send those separately by email or Drive.
- **The snapshot freezes at the moment you share.** Anything said afterwards stays private. To include later messages, unshare and share again — that regenerates the snapshot.
- **The three published pages are separate.** The submission protocol, output gallery, and model shortlist each have their own Share or Publish button on the page. Sharing the chat does not share them, and vice versa.

For a supervisor or co-author, I'd send the two .docx files directly and share the protocol page — those carry the substance. The chat link is mostly useful if you want them to see the reasoning behind the recommendations.

MD. ASHIQUR RAHMAN

8

Generate a pdf of the whole chat.

CLAUDE

This document.